Free-Boundary-Aware Risk Surfaces for American Options

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1 Introduction

American option risk surfaces, Greeks, and early-exercise boundaries.

2 Literature Review

Classical free-boundary methods, optimal stopping, and scientific machine learning.

3 Mathematical Formulation

Variational inequality and LCP formulation.

4 Benchmark Method

Crank–Nicolson, PSOR, boundary extraction, Greeks, and diagnostics.

5 Application Experiments

Stress-regime risk maps for American puts and dividend-paying American calls.

6 Neural Surrogate Workflow

Positive-premium price surrogate, direct boundary diagnostic head, bounded supervised Delta head.

7 Results

Price, boundary, Delta, violation, and runtime comparisons.

8 Limitations and Future Work

PSOR remains the audit standard. Extensions include larger regimes, active learning, uncertainty quantification, and richer dynamics.